University of the Philippines Diliman
College of Science
National Institute of Physics

Physics 180 THV-1
Problem Set 1-2

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IV - BS Physics

August 20, 2026

Problem 1.1

Go to `arXiv` and have a look at the relevant subjects for this course (nucl-ex, nucl-th, hep-ex, hep-ph, hep-th).

1. What do the first four digits of the `arXiv` number unique to a specific paper stand for?

2. What are the `arXiv` numbers for the papers shown in Slides 22 and 23.

3. Cite a single paper that fascinates you within the dates **August 11-20, 2026**, and briefly explain why you find this particular paper interesting.

Solution

1. The first four numbers in an `arXiv` paper stand for the year and month that the paper is published. The numbering system goes like

`arXiv:YYMM:NNNNN`

where `YY` is the last two digits of the year, and `MM` is the month of the submission. As such, majority of the recent `arXiv` submissions in the time of writing has the identifier code `arXiv.2608.NNNNN`. The rest of the numbers identify the paper within that month's submission sequence.

2. The paper shown in slides 22 and 23 have the following `arXiv` identifier numbers:

- ATLAS Run 2 searches for electroweak production of supersymmetric particles interpreted within the pMSSM

  `arXiv.2402.01392`(hep-ex)

- Multi-Higgs boson probes of the dark sector

  `arXiv.1912.02204`(hep-ph)

- Hitting two BSM particles with one lepton-jet: Search for a top partner decaying to a dark photon, resulting in a lepton-jet.

  `arXiv.2112.08425`(hep-ph)

3. A particular paper that I found interesting is `arXiv.2608.13206`(gr-qc)[1], entitled "Long-term 3+1 simulations of primordial black hole formation during radiation domination" by Ning et al. This interests me, well, because my field primarily right now is GR and I'm studying under Dr. Vega, so papers regarding general relativity and cosmology are of high interest to me. Reading the paper, they deal with the mathematics of primordial blackholes in 3+1 dimensions, and simulates the evolution of the collapse of such black holes during the early universe. and how it behaves near-critical collaps and how it evolves post-formation, and explicitly studies non-spherical symmetry. Very interesting indeed. I myself believe that primordial black holes were abundant in the early universe and is possibly one of the contributors of dark matter, so this paper is right up my alley. It's pretty neat!

Problem 1.2

Go to `InSpire` and search for all papers from 1967. What is the most highly-cited paper from that year? Skim through its introduction. Why do you think is it highly cited?

Solution

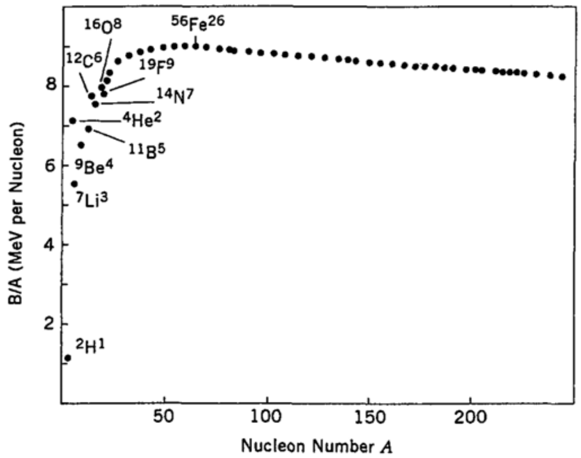
The most cited paper in `InSpire-HEP` in 1967 is "A Model of Leptons" by Steven Weinberg [2], which is also one of the most cited papers in physics of all time. The paper's influence appears to be from its groundbreaking proposal that leptons can be described by gauge fields and that bosons can acquire mass through spontaneous symmetry breaking, which is a revolutionary concept at the time. Skimming through the introduction, it describes a model in which the symmetry between the electromagnetic and weak interactions is spontenaeously broken, but Goldstone bosons (which is massless) are avoided. It introduces the photon and intermediate boson fields as gauge fields.

Looking past that, and reading even more papers about it, this paper is probably widely cited because it introduces an elegant model which unifies the electromagnetic and weak forces into what we now call the electroweak force, which is one of the stepping stones towards a unified theory of nature. It also shows how the W boson can acquire mass, and this paper led to testable predictions regarding the masses of the W and Z bosons, the existence of weak neutral currents mediated by the Z boson, and also predicted the existence of a new particle which would give mass to other particles, later known as the Higgs boson. These predictions are then confirmed in the following decades. And once the theory has been proven to be renormalizable (which the paper at its conclusion, wasn't really sure about), and once it was known to be compatible with the theory of the strong force, it became the foundation of the standard model.

Revolutionary content in just three pages. Insane.

Problem 2.1

The B/A plot looks like it peaks at a specific value of A. Using the curve approximated by SEMF, find the value of A that achieves this and find the peak value. What is the element that achieves this?



Solution

From [3], we have the Bethe-Weizsäcker or SEMF formula.

$$B = a_1 A - a_2 A^{2/3} - a_3 \frac{Z^2}{A^{1/3}} - a_4 \frac{(Z - A/2)^2}{A} \pm a_5 A^{-1/2} \tag{3.1}$$

Dividing by A, we have

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{Z^2}{A^{4/3}} - a_4 \frac{(Z - A/2)^2}{A^2} \pm a_5 A^{-3/2} \tag{3.2}$$

We aim to maximize B/A with respect to A. Note, however that the function is both a function of A and Z, so the resulting equation will be dependent of Z. An important point here is that Z is dependent of A. Since we wish to know which isobar A that an isotope Z gives the most stable configuration, we must first maximize B/A with respect to Z. This is similar to tbe operation performed inclass about finding N/Z, but we find the stability ratio for Z/A multiplied by A.

We find $Z(A)$ for which Z gives the most tightly bound isobar. Ignoring the pairing term for the smooth SEMF curve, we have

$$\frac{\partial B(Z,A)}{\partial Z}) = 0 \tag{3.3}$$

Therefore, we have

$$\frac{\partial}{\partial Z} B = \frac{\partial}{\partial Z}\left(a_1 A - a_2 A^{2/3} - a_3 \frac{Z^2}{A^{1/3}} - a_4 \frac{(Z - A/2)^2}{A} \pm a_5 A^{-1/2} \right) \tag{3.4}$$

$$\frac{\partial}{\partial Z} B = -2a_3 \frac{Z}{A^{1/3}} - 2a_4 \frac{(Z - A/2)}{A} \tag{3.5}$$

$$0 = -2a_3 \frac{Z}{A^{1/3}} - 2a_4 \frac{(Z - A/2)}{A} \tag{3.6}$$

We multiply the expression with $-A/2$ to simplifiy

$$0 = a_3 Z A^{2/3} + a_4\left(Z - \frac{A}{2}\right) \tag{3.7}$$

$$0 = a_3 Z A^{2/3} + a_4 Z - \frac{a_4 A}{2} \tag{3.8}$$

$$Z(a_3 A^{2/3} + a_4) = \frac{a_4 A}{2} \tag{3.9}$$

Thus, we have the most stable Z value for a given A

$$Z(A) = \frac{a_4 A}{2(a_3 A^{2/3} + a_4)} \tag{3.10}$$

We substitute this back to B/A.

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{Z(A)^2}{A^{4/3}} - a_4 \frac{(Z(A) - A/2)^2}{A^2} \pm a_5 A^{-3/2} \tag{3.11}$$

from this, we get

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{\left(\frac{a_4 A}{2\left(a_3 A^{2/3} + a_4\right)}\right)^2}{A^{4/3}} - a_4 \frac{\left(\frac{a_4 A}{2\left(a_3 A^{2/3} + a_4\right)} - \frac{A}{2}\right)^2}{A^2} \pm a_5 A^{-3/2} \tag{3.12}$$

This simplifies to the following

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - a_3 \frac{a_4^2 A^{2/3}}{4(a_3 A^{2/3} + a_4)^2} - a_4 \frac{a_3^2 A^{4/3}}{4(a_3 A^{2/3} + a_4)^2} \pm a_5 A^{-3/2} \tag{3.13}$$

Combining the a_3 and a_4 terms, we have

$$-a_3 \frac{a_4^2 A^{2/3}}{4(a_3 A^2/3 + a_4)^2} - a_4 \frac{a_3^2 A^{4/3}}{4(a_3 A^2/3 + a_4)^2} = -\frac{a_3 a_4 A^{2/3}(a_4 + a_3 A^{2/3})}{4(a_3 A^{2/3} + a_4)^2} \tag{3.14}$$

$$= -\frac{a_3 a_4 A^{2/3}}{4(a_3 A^{2/3} + a_4)} \tag{3.15}$$

Hence,

$$\frac{B}{A} = a_1 - a_2 A^{-1/3} - \frac{a_3 a_4 A^{2/3}}{4(a_3 A^{2/3} + a_4)} \pm a_5 A^{-3/2} \tag{3.16}$$

Now, we maximize this in terms of A. We have[1]

$$\frac{\partial}{\partial A}\left(\frac{B}{A}\right) = \frac{a_2}{3} A^{-4/3} - \frac{a_3 a_4}{4} \frac{\partial}{\partial A}\left[\frac{A^{2/3}}{a_3 A^{2/3} + a_4}\right] \mp \frac{3a_5}{2} A^{-5/2} \tag{3.17}$$

$$= \frac{a_2}{3} A^{-4/3} - \frac{a_3 a_4^2}{6} \frac{A^{-1/3}}{(a_3 A^{2/3} + a_4)^2} \mp \frac{3a_5}{2} A^{-5/2} \tag{3.18}$$

This maximum satisfies

$$\frac{a_2}{3} A^{-4/3} - \frac{a_3 a_4^2}{6} \frac{A^{-1/3}}{(a_3 A^{2/3} + a_4)^2} \mp \frac{3a_5}{2} A^{-5/2} = 0 \tag{3.19}$$

[1] At this point I just used mathematica to solve for this shit

For a smooth SEMF curve, we usually drop the pairing term, giving us

$$\frac{a_2}{3}A^{-4/3} - \frac{a_3 a_4^2}{6}\frac{A^{-1/3}}{(a_3 A^{2/3} + a_4)^2} = 0 \tag{3.20}$$

This can then be further simplified by multiplying by $6A^{4/3}$

$$2a_2 - \frac{a_3 a_4^2 A}{(a_3 A^{2/3} + a_4)^2} = 0 \tag{3.21}$$

Therefore

$$2a_2(a_3 A^{2/3} + a_4)^2 = a_3 a_4^2 A \tag{3.22}$$

If we solve for A, we end up with this quartic equation if $A = x^3$, basically a fractional power. This has no trivial closed form analytic solution.

$$2a_2 a_3^2 A^{4/3} - a_3 a_4^2 A + 4a_2 a_3 a_4 A^{2/3} + 2a_2 a_4^2 = 0 \tag{3.23}$$

We solve for this numerically. I just used the Newton-Raphson zero finding algorithm in `C`. Code is provided on the appendices to show how that works (see Listing 1).

Anyways, the resulting number for stable A, using the provided coefficients[3]

$$\begin{aligned} a_1 &= 15.56\,\text{MeV} \\ a_2 &= 17.23\,\text{MeV} \\ a_3 &= 0.697\,\text{MeV} \\ a_4 &= 93.14\,\text{MeV} \\ a_5 &= 12\,\text{MeV} \end{aligned}$$

Numerically solving for A results to the value $A \approx 62$, with $B/A \approx 8.76\,\text{MeV}$. Using the derived equation (3.10), we can calculate the most stable Z value for this A, and this results to $Z \approx 28$, or, **Nickel-62**.

$$\boxed{\begin{aligned} N &\approx 62 \qquad (3.24) \\ Z &\approx 28 \qquad (3.25) \end{aligned}}$$

The relevant graph is provided in the appendices (see Fig. 1), compared with actual binding energy per nucleon [4]. While I expected something akin to $^{56}_{26}$Fe, turns out, when you derive the actual energies, $^{62}_{28}$Ni is also an answer, since it has the highest nuclear binding energy per nucleon. However, Iron-56 is more commonly called the most stable end-product of stellar nucleosynthesis because it has the lowest mass per nucleon and is produced in massive quantities by stars [5]

So hopefully this passes by the checker. Nickel and Iron are just beside each other on the periodic table. $^{62}_{28}$Ni is valid.

References

[1] Zhuan Ning, Rong-Gen Cai, Shao-Jiang Wang, and Chul-Moon Yoo. Long-term 3+1 simulations of primordial black hole formation during radiation domination. *arXiv preprint*, 2026. doi: 10.48550/arXiv.2608.13206. URL https://doi.org/10.48550/arXiv.2608.13206. Submitted on 13 Aug 2026. Report number: NU-QG-25.

[2] Steven Weinberg. A Model of Leptons. *Physical Review Letters*, 19:1264–1266, Nov 1967. doi: 10.1103/PhysRevLett.19.1264. URL https://doi.org/10.1103/PhysRevLett.19.1264. Submitted November 1967.

[3] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website, UVLé.

[4] Meng Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The AME 2020 atomic mass evaluation (II). Tables, graphs and references. *Chinese Physics C*, 45(3): 030003, 2021. doi: 10.1088/1674-1137/abddaf. URL https://doi.org/10.1088/1674-1137/abddaf.

[5] Christopher S. Baird. What is the most stable nucleus? *Science Questions with Surprising Answers*, July 2024. URL https://www.wtamu.edu/~cbaird/sq/mobile/2024/07/23/what-is-the-most-stable-nucleus/. Accessed: 16-08-2026.

Appendices

Graph

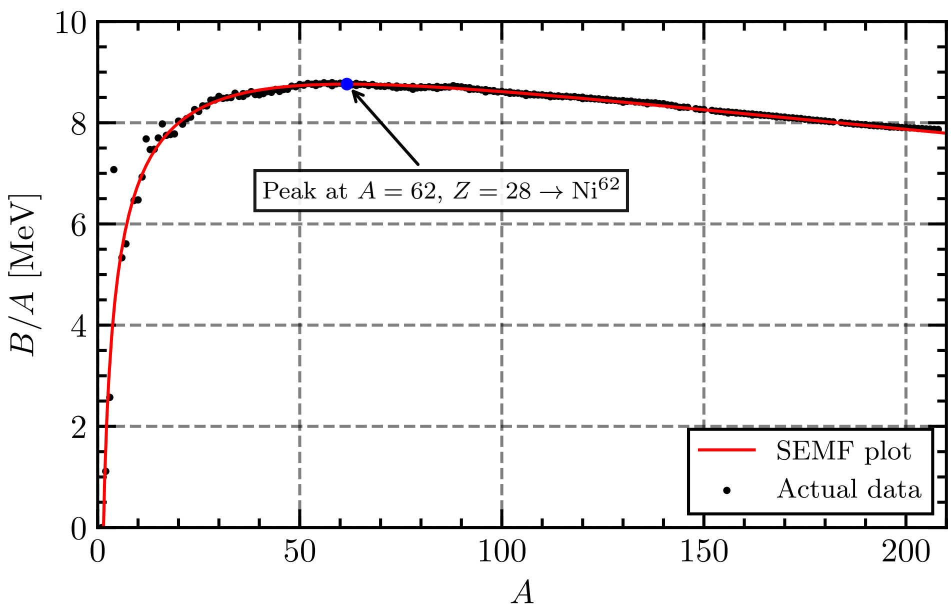


Figure 1: Binding energy per nucleon: SEMF vs. AME2020 data

Code

```c
#include <stdio.h>
#include <math.h>

// SEMF coefficients from [3]
double a1 = 15.56; // MeV
double a2 = 17.23;
double a3 =  0.697;
double a4 = 93.14;
double a5 = 12.00;

// Function definition f(A) - eq. 3.23
double f(double A) {
    double result =
          2 * a2 * pow(a3,2) * pow(A,4.0/3.0)
        - a3 * pow(a4,2) * A
        + 4 * a2 * a3 * a4 * pow(A,2.0/3.0)
        + 2 * a2 * pow(a4, 2);
    return result;
}

// Function derivative definition - fprime
double fp(double A) {
    double result =
          (8.0/3) * a2 * pow(a3,2) * pow(A, 1.0/3)
        - a3 * pow(a4,2)
        + (8.0/3) * a2 * a3 * a4 * pow(A, -1.0/3);
    return result;
}

// Most stable Z for A
double Z(double A) {
    double num = a4 * A;
    double den = 2 * (a3 * pow(A,2.0/3) + a4);
    return num / den;
}

// Newton Raphson algorithm
double newton_raphson (double A0, double tol, int max_iter)
{
    double A = A0;
    printf("Iteration 0: A = %.8f, f(A) = %12.5e\n", A, f(A));

    double f_val;  // function value
    double fp_val; // function derivative value
    double A_new;  // iterable var
    for (int i = 1; i <= max_iter; i++) {
        f_val = f(A);
        fp_val = fp(A);

        if (fabs(fp_val) < 1e-15) {
            printf("Derivative too small, method failure\n");
        }

        A_new = A - f_val / fp_val;
        printf("Iteration %d: A = %12.8f, f(A) = %12.5e\n", i, A_new, f(A_new));

        if (fabs(A_new - A) < tol) {
            printf("\nConverged after %d iterations.\n", i);
            return A_new;
        }

        A = A_new;
    }

    printf("\nMax iterations reeached. Final A = %.8f\n", A);
    return A;
}

void horzline(char ch,int len) {
    for (int i = 0; i < len; i++) {
```

```
        printf("%c",ch);
    }
    printf("\n");
}
// ===============================
// MAIN ROUTINE
// ===============================
int main()
{
    horzline('=',60);
    printf("Solving for A using NR method\n");
    horzline('=',60);

    double A0 = 100;
    printf("\nStarting with A0 = %.8f\n", A0);
    horzline('-',60);
    double solution = newton_raphson(A0, 1e-12, 100);
    printf("Maximal value for A: A = %.8f\n", solution);
    printf("Most stable element: Z = %.8f\n", Z(solution));
    horzline('=',60);
    return 0;
}
```

Listing 1: Newton-Raphson solver script for A and Z

Output:

```
============================================================
Solving for A using NR method
============================================================

Starting with A0 = 100.00000000
------------------------------------------------------------
Iteration 0: A = 100.00000000, f(A) = -2.01545e+05
Iteration 1: A =  61.97467171, f(A) = -1.61385e+03
Iteration 2: A =  61.66458097, f(A) = -1.72475e-01
Iteration 3: A =  61.66454782, f(A) = -1.97906e-09
Iteration 4: A =  61.66454782, f(A) =  0.00000e+00

Converged after 4 iterations.
Maximal value for A: A = 61.66454782
Most stable element: Z = 27.60762339
============================================================
```